

Project Development phase - 1

Preprocessing Steps and Business Questions with visualization

Date	8 July 2026
Team ID	SWTID-2026-5839
Project Name	Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau
Maximum Marks	

1. Data Preprocessing Steps

Step 1: Import the Dataset

- Imported the housing dataset into Tableau.
- Verified that all fields (Sale Price, Bedrooms, Bathrooms, Flat Area, etc.) were loaded correctly.

Step 2: Check for Missing Values

- Examined the dataset for null or blank values.
- Missing values were removed or handled appropriately to avoid incorrect analysis.

Step 3: Remove Duplicate Records

- Checked for duplicate house records.
- Removed duplicate entries to ensure every house was counted only once.

Step 4: Validate Data Types

- Ensured:
 - Sale Price → Number
 - FlatArea → Number
 - Age of House → Number
 - Years Since Renovation → Number
 - Latitude & Longitude → Geographic fields
 - Renovation Status → Categorical field

Step 5:

Verify Data Consistency

- Checked for unrealistic values such as:
 - Negative sale prices of
 - Negative house age
 - Zero or invalid area values
- Corrected or removed inconsistent records.

Step 6: Prepare Calculated Fields

Created calculated fields where required, such as:

- Average Sale Price
- Count of Houses
- Total Sales by Renovation Years
- House Age Categories (if required) etc.

Step 7: Final Clean Dataset

After preprocessing, the dataset became ready for visualization and dashboard development in Tableau.

2. Business Question with Visualization

S.NO.	Business Question	Visualization
1.	How many housing records are available for analysis after data preprocessing?	Count of Transformed Housing Data
2.	What is the average sale price of houses in the dataset?	Average Sale Prices
3.	How is the basement area distributed among houses?	Area of House from Basement (in Sqft)
4.	How do house sales vary based on the number of years since renovation?	Total Sales by Years Since Renovation
5.	How does the age of houses differ between renovated and non-renovated properties?	Distribution of House Age by Renovation Status

6.	How is house age related to the number of bedrooms, bathrooms, and floors?	House Age Distribution by Number of Bathrooms, Bedrooms, and Floors
7.	How does the Living Area show Impact on Sale Price	Impact of Living Area on Sale Price
8.	How does condition of house and view points show impact on sales price	Average Sale Price by Condition and View

3. Visualizations:

Count of Transformed Housing Data

Average Sale Prices

Area of House from Basement (in Sqft)

Count of
Transformed_
Housing_Data

4,205

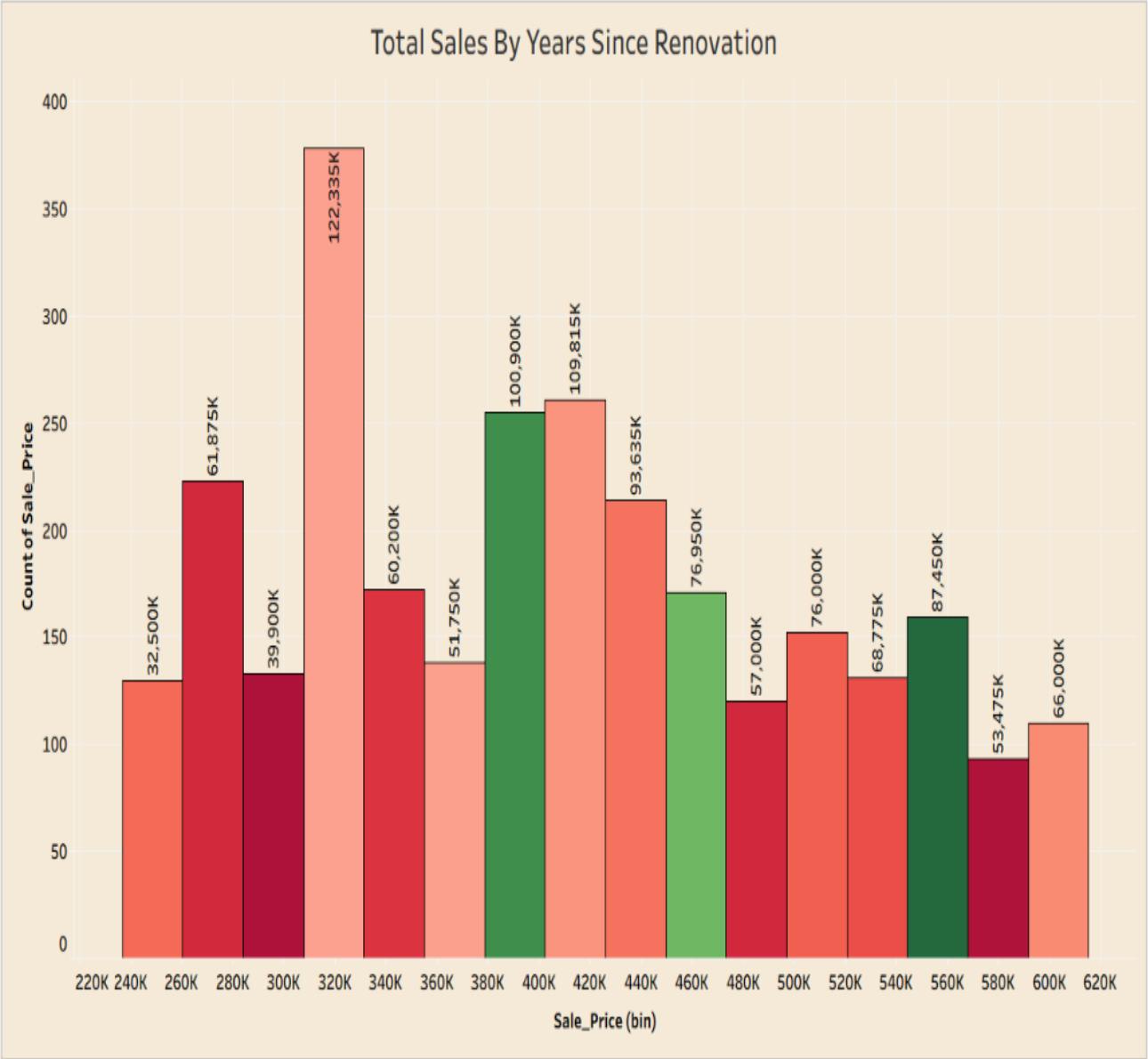
Average Sales Prices

619,759

Area of the House
from Basement (in
Sqft)

8,516,262

Total Sales by Years Since Renovation



Sale_Price

Limit

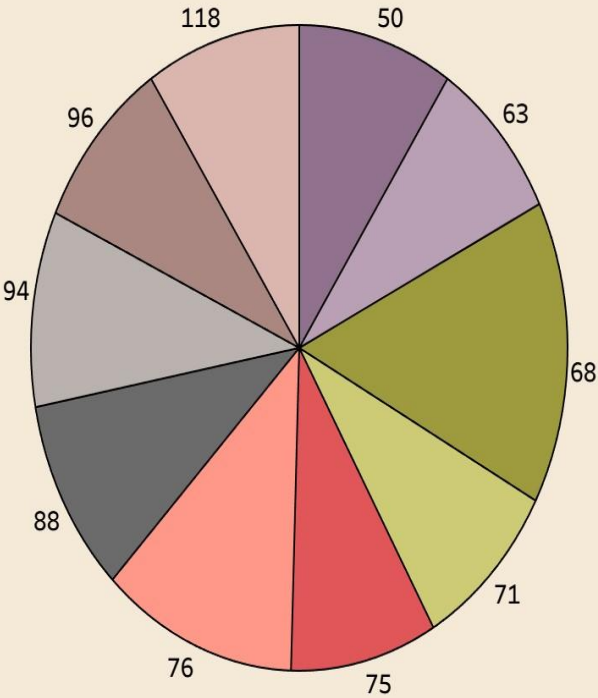
Top 25 by SUM([Age of House (in Years)])

SUM(Years Since Renovation)



Distribution of House Age by Renovation Status

Distribution of House Age by Renovation Status



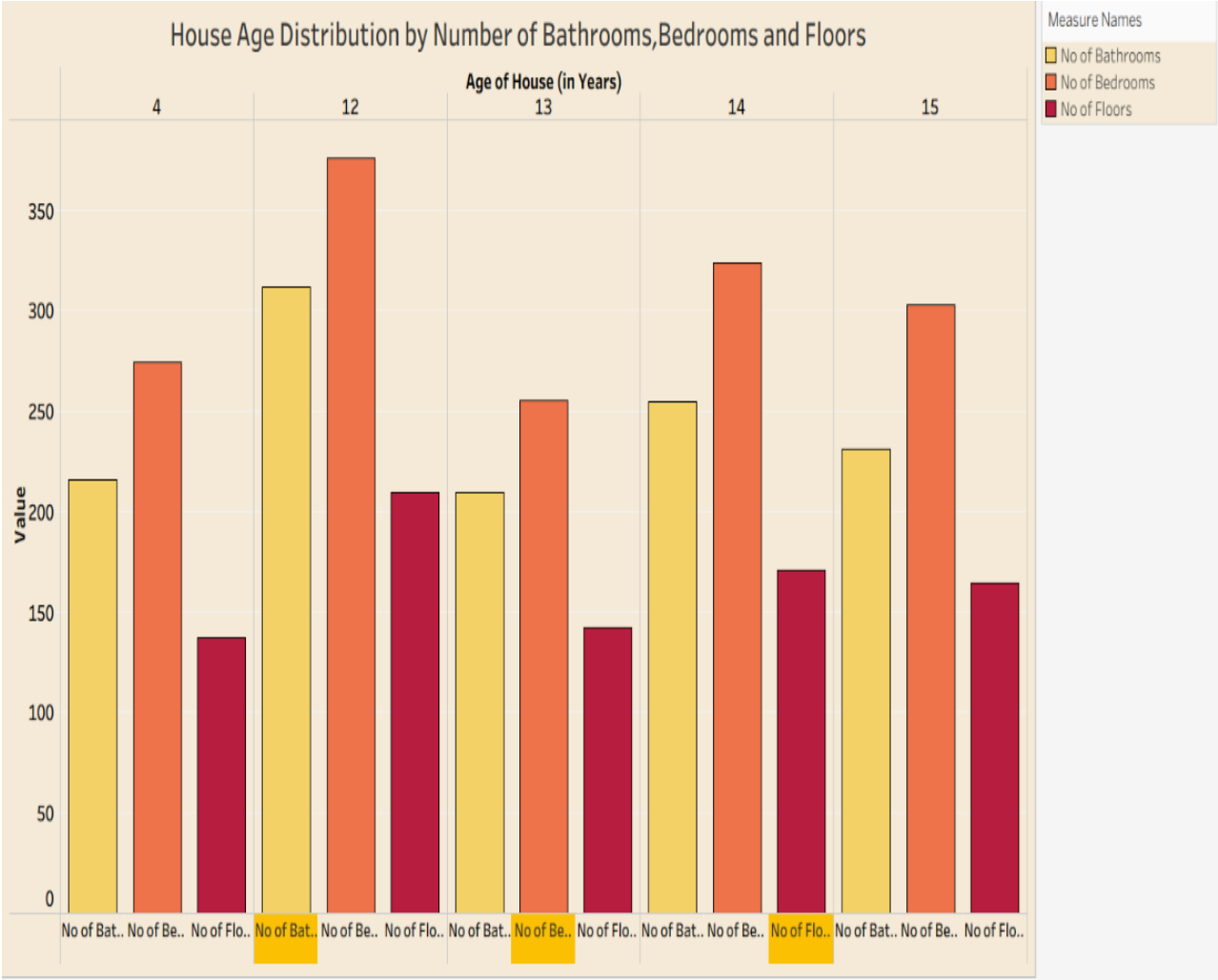
Age of House (in Years)

- 50
- 63
- 68
- 71
- 75
- 76
- 88
- 94
- 96
- 118

SUM(Ever Renovated Y...

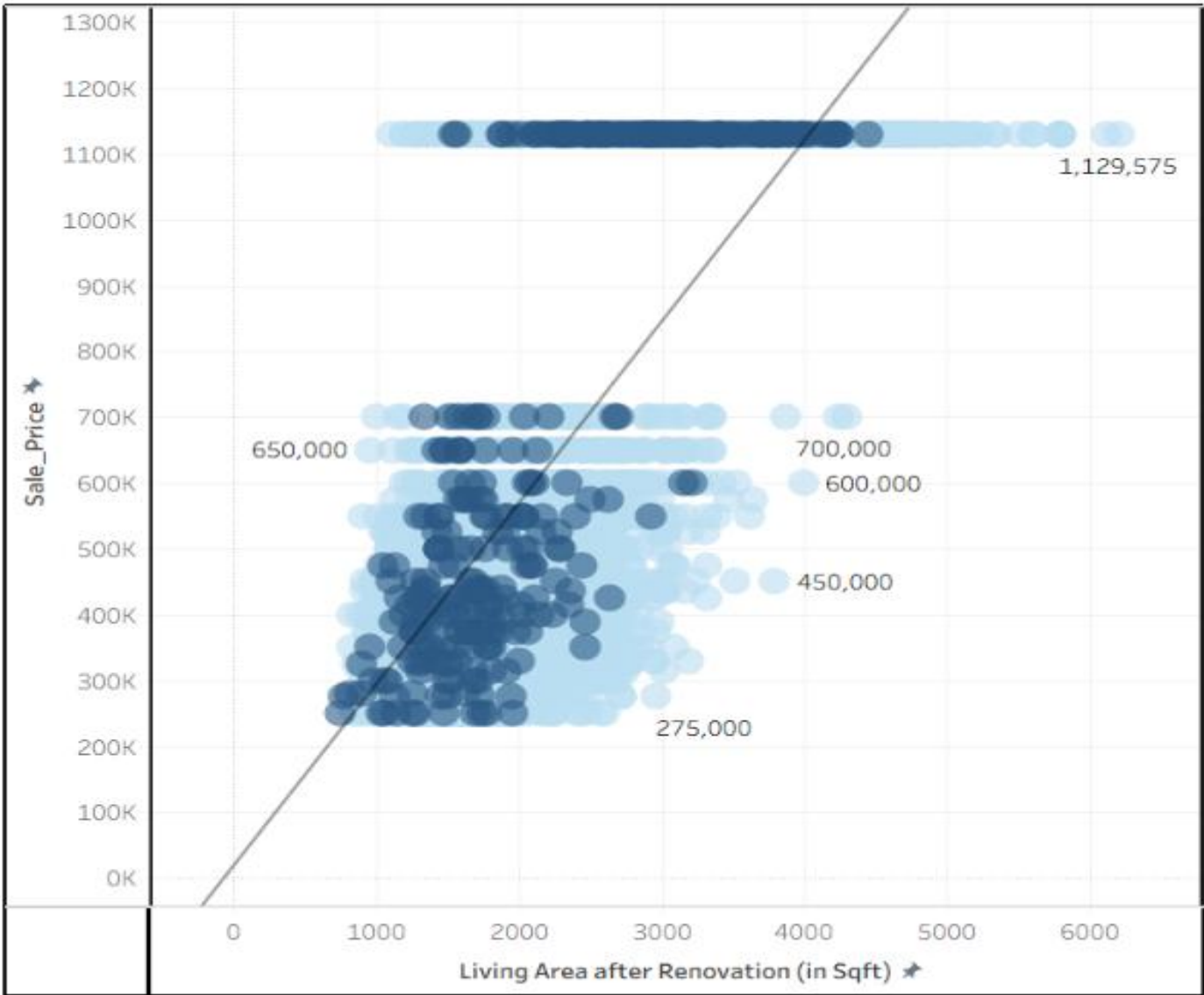
204

House Age Distribution by Number of Bathrooms, Bedrooms, and Floors

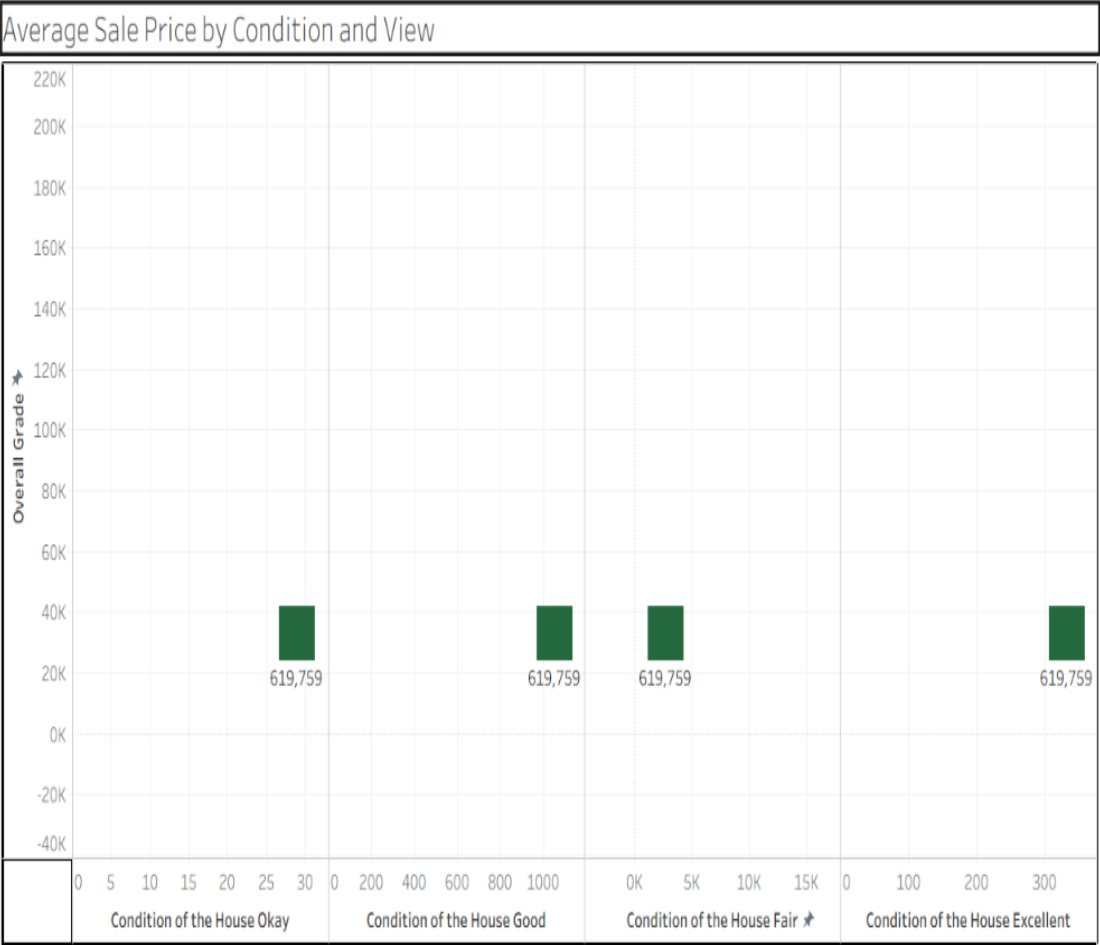


Impact of Living Area on Sale Price

Impact of Living Area on Sale Price



Average Sale Price by Condition and View



AVG(Sale_Price)

619,759